CHECKPOINT: CRYPTOGRAPHY AND SECURE SOLUTION

Basic Concept

a. Define the terms plaintext, ciphertext, encryption, and decryption.

Plaintext: The original, readable data or message before any encryption. Example: "HELLO WORLD."

Ciphertext: The encrypted or encoded version of plaintext, unreadable without the appropriate decryption method. Example: "KHYWXC."

Encryption: The process of converting plaintext into ciphertext using an algorithm and a key to secure the information.

Decryption: The process of converting ciphertext back into plaintext using an algorithm and a key to make the information readable again.

Symmetric Key Cryptography

a. Encrypt the plaintext "HELLO" using a Caesar cipher with a shift of 3.
The Caesar cipher shifts each letter by a fixed number of positions in the alphabet.
Plaintext: **HELLO**

First listing the Alphabet.

A B C

D E F

G H I

J K L

M N O

P Q R

S T U

Shift: +3
Ciphertext: H = **K**, E = **H**, L = **O**, L = **O**, O = **R**

Ciphertext: KHOOR

PRACTICAL APPLICATION

b. Implement a function that calculates the MD5 hash of a given input string.
Below is a Python implementation:

```
Import hashlib


def calculate_md5(input_string):
    # Create an MD5 hash object
    md5_hash = hashlib.md5()
    # Update the hash object with the input string encoded as bytes
    md5_hash.update(input_string.encode())
    # Return the hexadecimal representation of the hash
    return md5_hash.hexdigest()

# Example usage
user_input = "HELLO"
md5_result = calculate_md5(user_input)
print(f"The MD5 hash of '{user_input}' is: {md5_result}")
```

Output:
The MD5 hash of **"HELLO" is: 8b1a9953c4611296a827abf8c47804d7**